

NovaTech Screenshots - Data Transformation

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Covers the join diagram, join configuration, calculated fields, and data type corrections required by the Data Preparation and Quality rubric criteria.

Part A: SPICE Import Confirmation

Reviewer feedback requested screenshots confirming all three CSV datasets imported into SPICE successfully, with row counts and column counts matching the source files.

A1. SPICE Import Confirmation - crm_deals_raw

Amazon Quick

Datasets > crm_deals_raw

UdacityQuicksightLab

< Datasets

crm_deals_raw

Generate analysis

Chat

Edit dataset

Summary

Refresh

Permissions

Usage

Columns (21)

Search

Name	Type	Description
account_id	String	
company_name	String	
industry	String	
company_size_tier	String	
annual_revenue_usd	Decimal	
employee_count	Integer	
headquarters	String	
opportunity_id	String	
sales_rep	String	
sales_manager	String	
sales_region	String	
product_name	String	
product_category	String	
list_price	Integer	
deal_stage	String	
deal_created_date	Datetime	
deal_closed_date	Datetime	
deal_value	Decimal	
loss_reason	String	
data_source	String	

Items per page 20 < 1 2 >

About

SPICE Size: 270.5 KB

Refresh

Completed 499 rows imported (100% success)

Last refresh at September 14, 2026 at 11:24 PM GMT+8

Access

Row-level security Set up

Column-level security Set up

Sources

novatech_crm_deals.csv

Usage

Analyses (1)

Dashboards (1)

Datasets (0)

A2. SPICE Import Confirmation - marketing_campaigns_raw

Amazon Quick

Datasets > marketing_campaigns_raw

UdacityQuicksightLab

< Datasets

marketing_campaigns_raw

Generate analysis

Chat

Edit dataset

Summary

Refresh

Permissions

Usage

Columns (21)

Search

Name	Type	Description
account_id	String	
lead_id	String	
campaign_name	String	
campaign_channel	String	
campaign_date	Datetime	
funnel_stage	String	
customer_segment	String	
annual_income	Decimal	
household_size	Integer	
days_since_last_engagement	Integer	
product_spend_tier1	Integer	
product_spend_tier2	Integer	
product_spend_tier3	Integer	
campaign_spend	Decimal	
revenue_attributed	Decimal	
web_visits_per_month	Integer	
campaign_response	Integer	
Mkt_Src_Cd	String	
complaint_flag	Integer	
data_source	String	

Items per page 20 < 1 2 >

About

SPICE Size: 935.2 KB

Refresh

Completed 2240 rows imported (100% success)

Last refresh at September 14, 2026 at 9:13 PM GMT+8

Access

Row-level security Set up

Column-level security Set up

Sources

novatech_marketing_campaigns.csv

Usage

Analyses (1)

Dashboards (1)

Datasets (1)

A3. SPICE Import Confirmation - support_tickets_raw

Amazon Quick

Datasets > support_tickets_raw

UdacityQuicksightLab

Datasets

support_tickets_raw

Generate analysis

Chat

Edit dataset

Summary

Refresh

Permissions

Usage

Columns (21)

Search

Name	Type	Description
account_id	String	
ticket_id	String	
ticket_created_date	Datetime	
ticket_resolved_date	Datetime	
priority	String	
product_area	String	
contact_channel	String	
reported_by	String	
customer_tier	String	
company_size	String	
region	String	
users_affected	Integer	
error_rate_pct	Decimal	
downtime_minutes	Integer	
payment_impact	Integer	
security_incident	Integer	
data_loss	Integer	
customer_sentiment	String	
tickets_last_30_days	Integer	
data_source	String	

Items per page 20 < 1 2 >

About

SPICE Size: 1.3 MB

Refresh

Completed 3000 rows imported (100% success)

Last refresh at September 14, 2026 at 9:15 PM GMT+8

Access

Row-level security

Column-level security

Set up

Set up

Sources

novatech_support_tickets.csv

Usage

Analyses (0)

Dashboards (0)

Datasets (1)

Part B: Data Type Corrections and Calculated Fields

B1. Data Type Corrections

e.g., annual_revenue_usd, deal_value → Decimal; employee_count, list_price → Integer; deal_created_date, deal_closed_date → Date

Amazon Quick

Datasets > crm_deals_raw

UdacityQuicksightLab

File

Data

Help

SPICE

Save & publish

Cancel

TRANSFORM

OUTPUT

DATASET: 89

Switch back to legacy experience

Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

Add calculated columns

Change data type

Rename columns

Select columns

COMBINE TRANSFORMATIONS

Append

Join

OTHER TRANSFORMATIONS

Aggregate

Filter

Pivot

Unpivot

novatech_crm_deals.csv

Change data type 1

Add calculated...

Configure

Preview

Change data type 1

Select columns to change data type

Search columns

account_id

company_name

industry

company_size_tier

annual_revenue_usd

employee_count

headquarters

opportunity_id

sales_rep

sales_manager

sales_region

product_name

product_category

list_price

deal_stage

deal_created_date

deal_closed_date

deal_value

loss_reason

data_source

6 columns with data type changed

COLUMN NAME	ORIGINAL DATA TYPE	NEW DATA TYPE
annual_revenue_usd	Decimal	Decimal
employee_count	Integer	Integer
list_price	Integer	Integer
deal_created_date	Datetime	Date
deal_closed_date	Datetime	Date
deal_value	Decimal	Decimal

Select a column to change data type

B2. Calculated Field - days_to_close

Amazon Quick Datasets > crm_deals_raw

File Data Help

SPICE Save & publish Cancel

TRANSFORM OUTPUT DATASET: novatech_crm_deals.csv

Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

Add calculated columns

Change data type

Rename columns

Select columns

COMBINE TRANSFORMATIONS

Configure Preview

Add calculated columns 1

1 calculated column

+ Calculated column

Search columns

Column name	Calculation definition	Actions
# days_to_close	dateDiff(deal_created_date, deal_closed_date, "DD")	

days_to_close
Integer
124
105
14
111
128
80
12
8
7
20
125
127
59

B3. Calculated Field - campaign_roi

Amazon Quick Datasets > marketing_campaigns_raw

File Data Help

SPICE Save & publish Cancel

TRANSFORM OUTPUT DATASET: novatech_marketing_campaigns_raw

Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

Add calculated columns

Change data type

Rename columns

Select columns

COMBINE TRANSFORMATIONS

Append

Configure Preview

Add calculated columns 1

1 calculated column

+ Calculated column

Search columns

Column name	Calculation definition	Actions
# campaign_roi	(revenue_attributed - campaign_spend) / campaign_spend	

campaign_roi
Decimal
-0.7420536031297804
-0.6606384953288374
-1.0
-1.0
-0.9762840177664018
-0.5469012849499949
-0.7689286905928722
-1.0
-1.0
-1.0
-1.0
-1.0
-0.5657858418646726

B4. Calculated Field - resolution_time_hours

Amazon Quick | Datasets > support_tickets_raw

File Data Help

SPICE Save & publish Cancel

TRANSFORM OUTPUT DATASET: Switch back to legacy experience

Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

Add calculated columns

Change data type

Rename columns

Select columns

COMBINE TRANSFORMATIONS

Append

Join

OTHER TRANSFORMATIONS

Aggregate

Filter

novatech_support_tickets... → Change data type 1 → Add calculated...

Configure Preview

Add calculated columns 1

1 calculated column

+ Calculated column

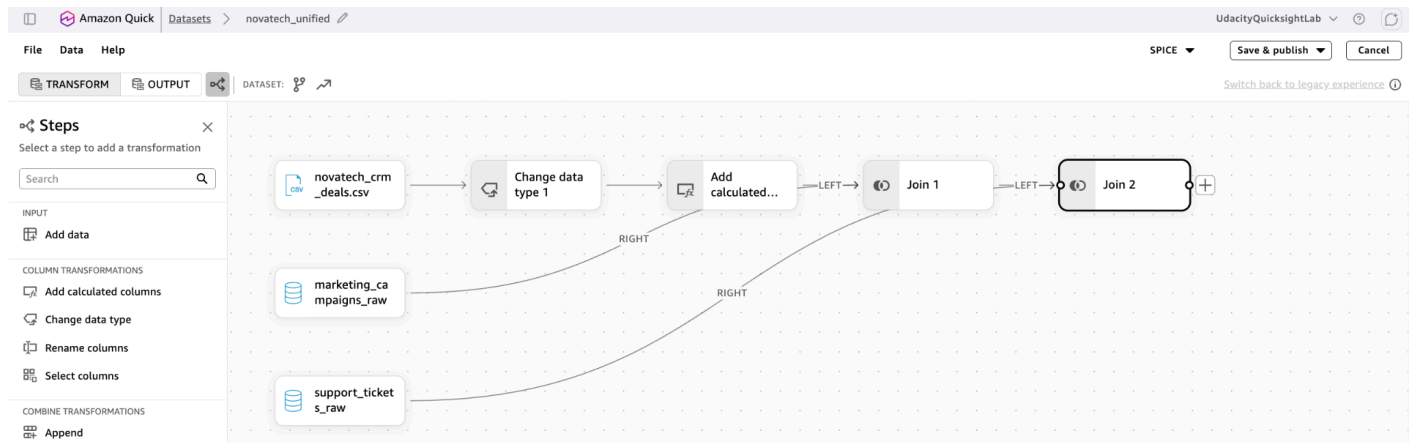
Search columns

Column name	Calculation definition	Actions
# resolution_time_hours	dateDiff(ticket_created_date, ticket_resolved_date, "HH")	

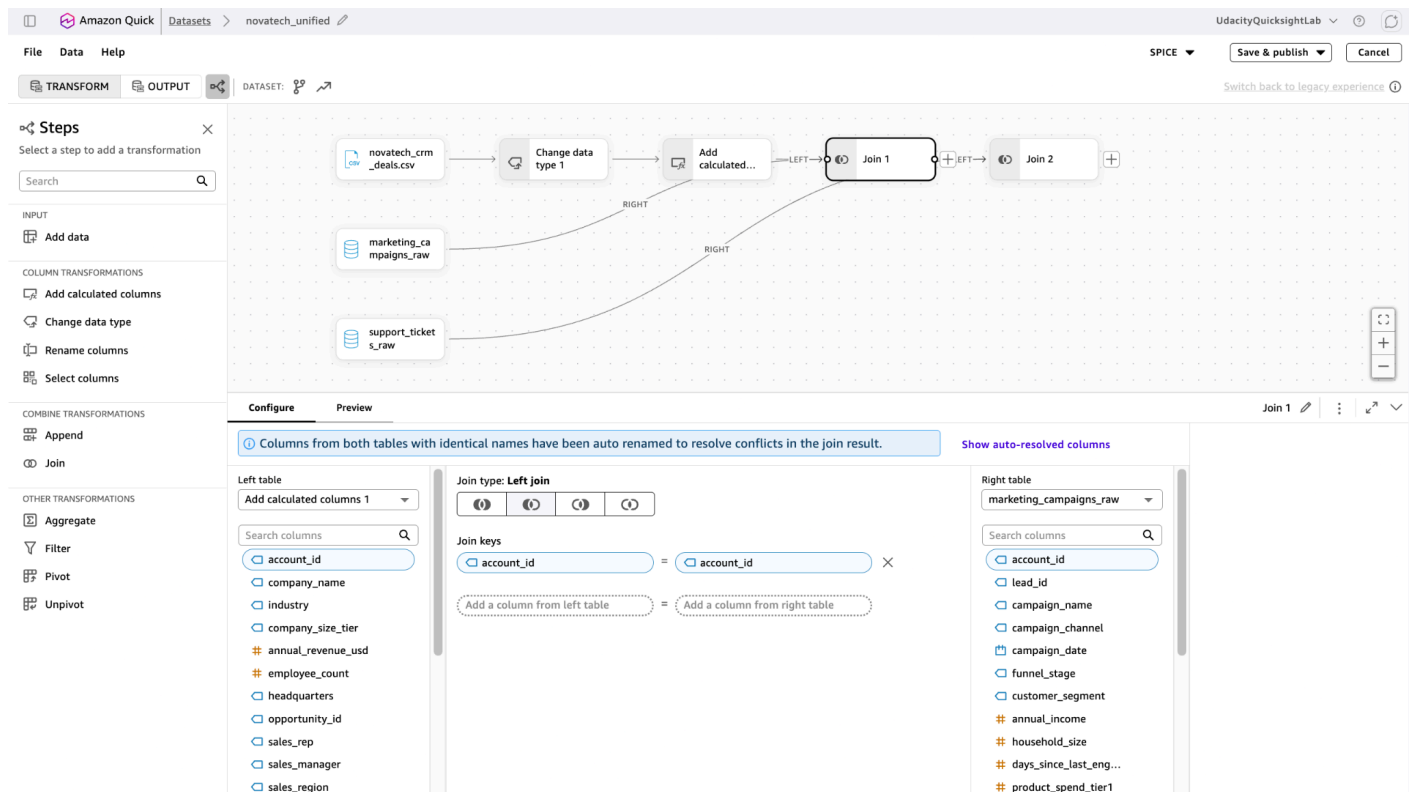
resolution_time_hours
Integer
7
93
113
131
103
64
14
35
58

Part C: Unified Dataset Join

C1. Join Diagram



C2. Join Configuration



Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

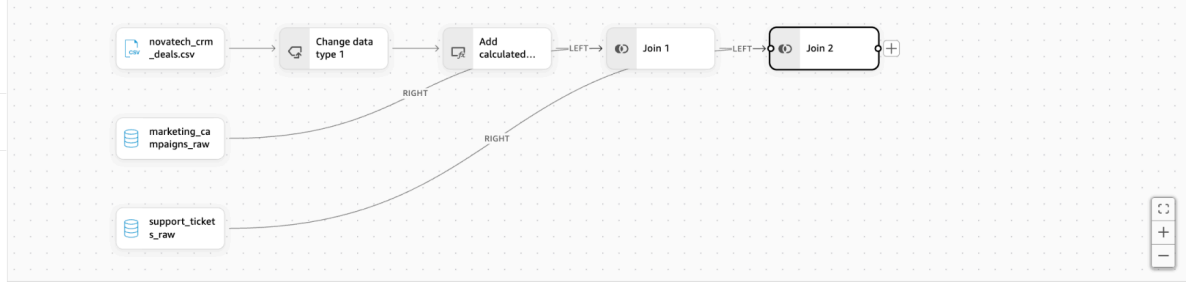
- Add calculated columns
- Change data type
- Rename columns
- Select columns

COMBINE TRANSFORMATIONS

- Append
- Join

OTHER TRANSFORMATIONS

- Aggregate
- Filter
- Pivot
- Unpivot



Configure Preview

Join 2

Left table

Join 1

Search columns

- account_id.1
- company_name
- industry
- company_size_tier
- annual_revenue_usd
- employee_count
- headquarters
- opportunity_id
- sales_rep
- sales_manager
- sales_region
- product_name
- product_category

Join type: Left join

account_id.1

 =

account_id

Add a column from left table = Add a column from right table

Right table

support_tickets_raw

Search columns

- account_id
- ticket_id
- ticket_created_date
- ticket_resolved_date
- priority
- product_area
- contact_channel
- reported_by
- customer_tier
- company_size
- region
- users_affected
- error_rate_pct